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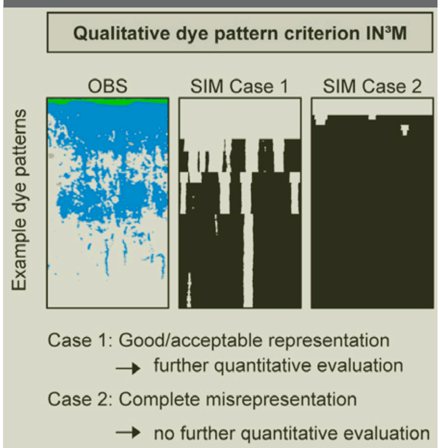
Benchmarking of Two Dual-Permeability Models under Different Land Use and Land Cover

Land use and land cover (LULC) exert strong controls on soil properties and thus water flow in soils. The question arises whether existing models designed for simulating water and solute movement in heterogeneous soil include the effects of LULC on infiltration and percolation. To answer this question, the predictive capacity of two solute transport models was benchmarked to simulate flow processes in soils developed under different LULCs. To benchmark the model performance, dye tracer sprinkling experiments were conducted on five sites displaying similar soil texture and parent material but differences in LULC (two farmland sites, tilled and untilled, two grassland sites, and one deciduous forest site). Water content changes were continuously measured at different depths with time domain reflectometers during 60-mm irrigation over 4 h. After the tracer application, vertical and horizontal soil sections were excavated and photographed. The experimental data were then compared with the simulations based on a multicriteria benchmarking strategy, including simulated vs. observed water content changes, solute distribution profiles, and maximum infiltration depth. Both models were capable of predicting water flow for one of the grassland sites but revealed severe weaknesses when additional flow processes not explicitly included in the model structure came into play. Poor model outcomes resulted from LULC effects such as a strong surface microtopography altering macropore flow initiation in the agricultural soils and from horizontally oriented roots and surface water repellency in the forest soil. Our results suggest that LULC effects need to be better incorporated into the conceptualization and parameterization of infiltration and percolation in hydrologic models to obtain realistic predictions concerning water quality and quantity.

Abbreviations: LULC, land use and land cover; MAE, mean absolute error; MDA, macropore drainage area; ME, mean error; PTF, pedotransfer function; TDR, time domain reflectometry.



A benchmark study was conducted to assess the predictive capacity of two dual-permeability models for structured soil under different land use and land cover. The comparison of simulations with experimental data from dye tracer experiments (dye pattern analysis and TDR measurements) revealed ambiguous results concerning the predictive capacity.

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Predicting the effect of land use change on water resources is of growing interest, especially in the context of climate change. Many recent studies have shown that LULC exert strong controls on soil properties and thus water flow in soils (e.g., Schwartz et al., 2003; Bodhinayake and Si, 2004; Bormann and Klaassen, 2008; Zhou et al., 2008; Bachmair et al., 2009; Wahren et al., 2009). Thus accurate model predictions of water flow and solute transport in soils under different LULCs are needed, particularly because solute transport models are increasingly being applied to address practical problems (Vanderborght et al., 2005).

Much progress has been made in identifying the processes governing flow and transport in structured soil and their translation into numerical models. There is now consensus that preferential flow phenomena "are the rule rather than the exception" during infiltration and percolation of water and solutes through the soil matrix (Flury et al., 1994). Depending on the mechanisms triggering preferential flow, heterogeneous matrix flow and macropore flow can be discerned (Weiler and Naef, 2003a). Heterogeneous matrix flow covers unstable flow and funnel flow (Hendrickx and Flury, 2001). Unstable flow results from instabilities in the wetting front due to textural boundaries, water repellency, or air entrapment in the soil matrix and can break up into finger-shaped flow paths. Funnel flow occurs when water is laterally diverted or funneled as a result of less permeable zones such as impeding layers or stones (Hendrickx and Flury, 2001). Macropore flow depends on the presence of large structural pores. These pores can be interaggregate pores or cracks resulting from intrinsic soil properties responding to abiotic factors such as weathering, swelling–drying, and freezing–thawing processes. Additionally, external factors such as bioturbation by earthworms and other soil organisms, root dynamics, and anthropogenic management practices like tillage and traffic cause structure development (Horn et al., 1994; Jarvis, 2007).

Soil texture, soil structure, topography, and surface microrelief have been identified as the primary factors governing preferential flow in soils. These factors constitute the potential for bypass flow. The extent to which this potential is realized depends on the initial and boundary conditions such as rainfall intensity, amount, and duration as well as antecedent soil moisture content (Jarvis, 2007). This already complex system of different interacting factors becomes more complex still when LULC are considered because vegetation and management practices alter many of these factors (e.g., Shipitalo et al., 2000; Bodhinayake and Si, 2004; Zhou et al., 2008). For instance, Bormann and Klaassen (2008) detected systematic impacts of land use on bulk density and soil hydraulic parameters by comparing cropland, grassland, and forest soils. Wahren et al. (2009) recently documented an increase in the number of medium and large pores in soils under afforestation compared with agricultural soil. Bachmair et al. (2009) also discovered significant distinctions in macropore structure, surface characteristics, and flow processes among sites with differing land uses.

The question arises whether the conceptualized and parameterized mechanisms governing infiltration and percolation in current, physically based solute transport models adequately include the effects of LULC (we use the term *solute transport model* for models simulating water flow and solute transport). One way to test the inherent capability of models to describe field data is through "blind" or "predictive" simulations (Ewen and Parkin, 1996), i.e., simulating experimental data using information about the study site without knowledge of the output data found at the site (Dubus et al., 2003). In consequence, the modeler does not have access to the output data for the test site and hence cannot calibrate the model (Ewen and Parkin, 1996). Because hydrologic models are often used as tools for scenario analyses, output data are not available (e.g., impact assessment of climate or land use change). Hence, "blind" simulations for cases with known output data are crucial for assessing the transferability of models to cases where conditions shift beyond our range of prior experience (e.g., due to extreme precipitation events, climate, or land use change) (Bathurst et al., 2004; Kirchner, 2006). In other words, we need to assure that we get "the right answers for the right reasons" in our models (Klemeš, 1986). Model calibration may lead to "the right answer," yet possibly because of "the wrong reason." Subsequently, the adequacy of the conceptualized and parameterized internal processes of current models needs to be examined by predictive simulations.

The objective of this study was to benchmark the predictive capacity of two dual-permeability models to correctly simulate flow and transport in structured soil developed under different LULCs. Two models were benchmarked: (i) the INfiltration–INitiation–INteraction model (IN^{3}M) (Weiler, 2001), and (ii) the well-known MACRO model, version 5.1 (Jarvis et al., 1991). The IN^{3}M model is a dual-permeability model developed for grassland soil containing earthworm channels, which has shown good agreement with field observations. The MACRO model is one of the most widely used dual-permeability models to date, used to support authorities in making decisions about the approval of pesticide registration in the European Union (Dubus and Brown, 2002; Scorza et al., 2007). Benchmark tests serve to "assess the performance of a code to either confirm its adequacy, capability and trustworthiness or to find out imperfections, limitations or inadequacies" (Oswald and Kinzelbach, 2004). Through predefined benchmarks of multiple data sets, the adequacy of model simulations with regard to different internal processes can be evaluated. Based on a benchmark test, we investigated the following research questions: Can the observed processes be adequately simulated with the current model structure of IN^{3}M and MACRO? Is it sufficient to simply account for land use effects by altering parameters linked to land use (e.g., different macroporosity per land use type)? How do land use and land cover modify the conceptualized processes in IN^{3}M and MACRO?

Since neither model was calibrated, the uncertainty associated with the derivation of input parameters as one source of model uncertainty has to be considered (Beven, 2001; Dubus et al., 2003). To account for this uncertainty, different ways of parameterizing IN^{3}M and MACRO were tested. For many practical applications, an inexpensive and rapid determination of soil hydraulic properties via pedotransfer functions (PTFs) is the only possible approach. To benchmark the model performance under these conditions, we decided to use PTFs for deriving the matrix hydraulic properties. Since both models are sensitive to the choice of matrix hydraulic properties, different PTFs were explored (Weiler, 2001; Dubus and Brown, 2002).

Materials and Methods

Field Experiments

As a first step toward benchmarking the predictive power of MACRO and IN^{3}M, dye tracer experiments were conducted on five sites displaying similar soil texture and parent material but with differences in LULC. Details on the study sites and field experiments are given elsewhere (Bachmair et al., 2009) and will here only be briefly summarized.

Two farmland sites (tilled and untilled), two grassland sites, and one deciduous forest site were investigated. All sites were located in northeastern Germany: the two farmland sites and one grassland site were situated on the outskirts of Berlin (52°28′ N, 13°18′ E), whereas the second grassland site and the forest site were situated in Brandenburg (52°1′ N, 12°36′ E). Due to their close spatial vicinity, soil texture and soil horizons were similar at the farmland and grassland sites in Berlin. Each was comprised of a silty sand in the topsoil and sandy loam in the subsoil. The grassland and forest sites in Brandenburg exhibited a higher proportion of silt than the Berlin sites (see Table 1 for soil properties for each site). All neighboring sites were spaced several meters apart to exclude lateral flow among them. The initial soil moisture content was found to be similar among the sites.

At each site, identical sprinkling experiments were conducted with a Brilliant Blue FCF solution of 4 g L^{-1} to visualize flow paths. The tracer experiments were performed in the fall of 2007 under weather conditions of low evaporation. The grass at both grassland sites was cut short (2–3 cm) before the tracer application. At the forest site, the freshly fallen litter layer was removed. Both farmland sites had bare soil. The tilled site was characterized by a ridge-and-furrow topography due to tillage operations. The microtopography of the untilled farmland site consisted of patchy depressions and hills. Harvesting had loosened the upper 10 to 15 cm of the untilled soil.

Table 1. Soil properties of the five experimental sites.

Site†	Soil series‡	Horizon‡	Depth	Particle size distribution§ Sand	Silt	Clay	Soil texture¶	Organic C#	Density††	pH ††
			cm	————	%	————		%	g cm^{-3}	
NT	Albic Luvisol	Ah	0–30	60.2	34.8	5.0	silty sand	1.9	1.0/1.2‡‡	6.2
		E	30–40	62.6	33.3	4.1	silty sand	0.7	1.5‡‡	6.3
		Bt-C	40–75/78	87.0	10.9	2.1	loamy sand	0.3	1.6‡‡	6.4
		C	75/78–100	50.0	41.2	8.8	sandy loam	0.8	1.7‡‡	6.2
T	Albic Luvisol	Ah	0–30	46.9	48.1	5.0	silty sand	2.7	1.0‡‡	5.8
		E_1	30–40	50.7	43.3	6.0	silty sand	1.1	1.7‡‡	5.9
		E_2	40–63	60.5	33.0	6.5	silty sand	0.5	1.8‡‡	5.9
		Bt-C	63–100	41.6§§	48.4§§	10.0§§	sandy loam	1.2§§	1.8‡‡	NA¶¶
G1	Albic Luvisol	Ah/E_1	0–38	54.2	38.7	7.1	silty sand	1.6	1.7‡‡	5.8
		E_2	38–50/70	57.9	35.1	7.0	silty sand	0.5	1.8‡‡	5.5
		Bt-C	50/70–100	41.6	48.4	10.0	sandy loam	1.2	1.8‡‡	5.8
G2	Gleyic Cambisol	Ah	0–20/22	31.5	64.1	4.4	sandy silt	3.1	1.2	4.5
		Bw_1	20/22–30/33	40.0	55.4	4.6	sandy silt	1.2	1.3	4.7
		Bw_2	30/33–53	75.1	22.5	2.4	loamy sand	0.5	1.3	4.6
		Bg/Cr	53–66	22.5	72.6	4.9	sandy silt	0.8	1.4	4.1
		Bg/Cr	66–75/80	62.1	34.6	3.3	silty sand	0.5	1.3	4.2
		Bg/Cr	75/80–100	30.8	63.7	5.5	sandy silt	1.2	1.4	4.0
F	Eutric Cambisol	L/Oh	+3–0							
		Ah	0–10	35.9	60.9	3.2	sandy silt	8.6	1.0	3.4
		Bw_1	10–15	31.6	63.1	5.3	sandy silt	2.4	1.1	3.8
		Bw_2	15–25	32.2	62.3	5.5	sandy silt	1.6	1.2§§	4.1
		BwC	25–30/55	35.6	59.1	5.3	sandy silt	1.5	1.2	4.1
		C	30/55–100	46.4	49.5	4.1	silty sand	0.5	1.1	4.3

† NT, untilled farmland; T, tilled farmland; G, grassland; F, forest.
‡ Soil classification from World Reference Base for Soils.
§ Sand: 2000–63 mm, silt: 63–2 mm, clay: <2 μm; measured with laser diffractometer.
¶ Soil texture classification after Sponagel (2005).
# By loss-on-ignition.
†† Average of two replicates.
‡‡ G1: data provided by the Faculty of Agriculture and Horticulture of Humboldt Univ., Berlin, Germany. No measurements were conducted at NT or T; due to the close spatial vicinity of NT, T, and G1, bulk density at NT and T was assumed to be similar to G1; bulk density for the tilled/harvested topsoil was estimated with data from the literature (Edwards et al., 1988): 1.0 for the Ap horizon (T) and 1.0 for 0–10 cm and 1.2 for 10–30 cm (NT).
§§ Data not available and hence estimated from adjacent horizons.
¶¶ NA = not available (missing data).

A high-intensity rainfall event was simulated at each site by sprinkling 60 mm of dyed water on a 50- by 100-cm area (15 mm h^{-1} for 4 h). Before, during, and after the sprinkling, 10 time domain reflectometry (TDR) probes measured the water content changes every 2 min in 10-, 20-, 30-, 40-, and 60-cm depths. The TDR probes had been previously mounted horizontally in the profile from a trench. One day after the irrigation, several vertical and horizontal soil sections were excavated and photographed. The sprinkled area was therefore divided into one half used for vertical sections and one half for horizontal sections. The pictures were processed using digital image analysis and the resulting dye patterns analyzed for several parameters. As relevant parameters for the model benchmark test, the mean volume density per depth increment of 1 mm as well as the maximum infiltration depth were calculated from the vertical dye patterns. The parameter volume density is similar to the frequently used dye coverage (e.g., Flury et al., 1994; Öhrström et al., 2002; Kasteel et al., 2005; Bogner et al., 2008). It is defined as the stained volume divided by the total volume (Weiler and Flühler, 2004). The horizontal pictures were analyzed for number, radius, and area of macropores as well as the distance between macropores. Macropores refer to visible interaggregate pores, earthworm burrows, and roots (live and decayed).

Model Description

A comprehensive technical description of both infiltration models can be found elsewhere, e.g., Weiler (2005) for IN^3M and Larsbo et al. (2005) or Scorza et al. (2007) for MACRO 5.1. A brief comparison is given below.

Both models are physically based, one-dimensional, dual-permeability models allowing for water flow both in the soil matrix and within discrete macropores. Generally, dual-permeability models divide the total pore space into one part characterized by a large storage capacity and small flow capacity (the soil matrix) and another part with a small storage capacity but large flow capacity (macropores) (Larsbo et al., 2005). Despite these broad commonalities, IN^3M and MACRO differ in the description of vertical flow and transport in both flow domains, lateral exchange between both domains (interaction), and assumptions about macropores. The MACRO model couples classical treatments of flow and transport processes in the matrix (Richards' equation, convection–dispersion equation) with non-capillary, gravity-driven macropore flow. The matrix and macropore domains are partitioned by a user-defined pressure head, Ψ_b. The water content θ_b and the hydraulic conductivity K_b represent the saturated state of the soil matrix (Larsson and Jarvis, 1999; Larsbo et al., 2005). Lateral water exchange between micropores and macropores is calculated as a function of an effective aggregate half-width, using an approximate, physically based, first-order function that neglects the influence of gravity and assumes a rectangular slab geometry for the aggregates (Scorza et al., 2007). In contrast, IN^{3}M is based on analytical solutions of the Green–Ampt equation to calculate both vertical and horizontal infiltration (Weiler, 2005). The interaction process in IN^{3}M is a function of the unsaturated hydraulic conductivity and the saturation deficit of the soil matrix (Weiler, 2005). Macropores are assumed to be vertically oriented, circular tubes.

What sets IN^{3}M apart from most infiltration models to date is the treatment of macropore flow initiation based on distributions of the macropore drainage area, which is defined as the individual microcatchment of each macropore. Generally, macropore flow is initiated once the inflow into a soil layer exceeds the infiltration capacity of the soil matrix. This can take place either at the soil surface or at the upper boundary of a subsurface layer. Because the distribution of flow rates in macropores was found to distinctly differ between surface and subsurface initiation, Weiler and Naef (2003b) derived distributions of the macropore drainage area (MDA) for surface and subsurface initiation. Thus, excess water is not evenly distributed into the macropores as in MACRO. Instead, macropore flow initiation is based on experimentally derived distributions of the macropore drainage area (Weiler and Naef, 2003b). Another special feature of IN^{3}M is a built-in module that generates artificial dye patterns of the simulated results, thereby providing the possibility of directly comparing simulated vs. experimental dye patterns (Weiler, 2001). The dye patterns are generated by simulating a macropore network in three dimensions based on the defined macropore parameters. The staining around each macropore is defined by the exchange between macropore and matrix (Weiler, 2005). The depth of the vertical infiltration front is used to define the depth of the homogeneous staining, assuming advection processes only.

Parameterization

The IN^{3}M model requires following input parameters: (i) matrix (micropore) properties—the number and thickness of soil layers, saturated hydraulic conductivity K_s, van Genuchten fitting parameters α and n, and saturated and residual moisture content θ_s and θ_r; (ii) macropore properties—the radius and number of macropores; and (iii) initial and boundary conditions—the duration and intensity of precipitation, initial saturation deficit, and maximum macropore outflow at the lower boundary.

The soil matrix was divided into 40 layers of 5-cm thickness. The matrix properties were estimated with two different PTFs: (i) the ROSETTA PTF (Schaap et al., 2001) based on the measured soil texture and bulk density; and (ii) the internal PTF of the MACRO model (required inputs: soil texture, bulk density, and organic matter content) (see Table 2). The parameters characterizing the macropore domain were derived from the horizontal soil section images (the macropore radius represents the median) (see Table

Table 2. Input parameters: soil micropore hydraulic properties† for IN^{3}M and MACRO estimated with the ROSETTA and MACRO pedotransfer functions (PTFs).

			ROSETTA PTF					MACRO PTF				
Site‡	Horizon	Depth	θ_s	θ_r	K_s	n	α	θ_s	θ_r	K_s	n	α
		cm	— cm^3 cm^{-3} —		mm h^{-1}		m^{-1}	— cm^3 cm^{-3} —		mm h^{-1}		m^{-1}
NT	Ah1	5–10	0.470	0.039	102.529	1.443	1.534	0.531	0.000	5.085	1.324	2.139
	Ah2	10–30	0.418	0.036	53.164	1.446	1.940	0.408	0.000	5.085	1.305	3.604
	E	30–40	0.356	0.031	21.547	1.428	3.459	0.381	0.000	5.719	1.375	3.570
	Bt-C	40–75	0.354	0.042	90.859	2.384	4.044	0.317	0.000	14.961	1.501	6.565
	C	75–100	0.313	0.032	5.488	1.326	2.861	0.337	0.000	3.088	1.255	1.981
T	Ah	0–30	0.448	0.042	86.764	1.580	0.661	0.529	0.000	3.119	1.285	2.313
	E1	30–40	0.307	0.029	6.909	1.324	3.284	0.338	0.000	3.472	1.288	1.611
	E2	40–65	0.294	0.029	5.743	1.277	5.616	0.306	0.000	4.896	1.268	2.060
	Bt-C	65–100	0.291	0.032	2.874	1.289	2.645	0.305	0.000	2.174	1.203	0.905
G1	Ah/E1	0–40	0.312	0.030	7.005	1.320	3.615	0.329	0.000	3.809	1.271	1.505
	E2	40–70	0.293	0.029	5.213	1.267	5.259	0.308	0.000	4.378	1.260	1.895
	Bt-C	70–100	0.291	0.032	2.874	1.289	2.645	0.305	0.000	2.166	1.203	0.901
G2	Ah	0–20	0.403	0.043	48.181	1.692	0.486	0.472	0.000	1.806	1.282	1.930
	Bw1	20–30	0.373	0.037	29.178	1.582	0.759	0.452	0.000	2.452	1.348	2.280
	Bw2	30–55	0.418	0.035	72.255	1.549	4.242	0.430	0.000	9.568	1.447	4.069
	Bg/Cr	55–65	0.382	0.044	22.105	1.656	0.593	0.432	0.000	1.277	1.385	1.302
	Bg/Cr	65–80	0.393	0.032	43.574	1.438	2.585	0.442	0.000	5.763	1.414	3.209
	Bg/Cr	80–100	0.365	0.040	20.204	1.619	0.662	0.428	0.000	1.698	1.341	1.704
F	Ah	0–10	0.447	0.044	106.662	1.704	0.427	0.531	0.000	2.208	1.176	3.017
	Bw1	10–15	0.430	0.047	66.677	1.725	0.418	0.509	0.000	1.767	1.286	2.364
	Bw2	15–25	0.404	0.044	44.197	1.692	0.481	0.482	0.000	1.788	1.318	2.156
	BwC	25–55	0.399	0.042	43.549	1.664	0.529	0.482	0.000	2.039	1.323	2.246
	C	55–100	0.418	0.038	62.108	1.568	0.741	0.516	0.000	3.154	1.417	1.769

† θ_s and θ_r, saturated and residual water contents, respectively; K_s, saturated hydraulic conductivity; n and α, fitted parameters.

‡ NT, untilled farmland; T, tilled farmland; G, grassland; F, forest.

3). The intensity and duration of precipitation were 15 mm h^{-1} for 4 h. The initial saturation deficit was calculated from the measured initial soil moisture and θ_s (see Table 3). For the maximum macropore outflow at the lower boundary, the K_s of the deepest layer was chosen. The two parameterization cases differed only in matrix properties.

For the parameterization of the MACRO model, we applied three different strategies. For Case A, the micropore and macropore parameters were estimated with the internal MACRO PTFs. For the calculation of the macropore parameters, the properties of the aggregate structure have to be specified (structure and strength and shape of aggregates according to the FAO soil structural description). We chose a strong aggregate structure because the experimental soil sections were interspersed with numerous structural features (biopores). For Cases B and C, the macropore parameters were derived from the experimental data (for the calculation of individual macropore parameters, see Table 4). Cases B and C differ in the method used to estimate the micropore parameters (Case B, the MACRO PTF; Case C, the ROSETTA PTF).

For meteorological conditions, we used estimated temperature data and precalculated evapotranspiration rates. As a bottom boundary condition, we specified a constant hydraulic gradient, and as initial conditions, user-defined water contents (TDR data). To limit the number of parameters, we simulated bare soil for all sites because specifying crop-related parameters only alters evapotranspiration and interception rates and not soil properties. As the solute, we chose a nonreactive tracer; the number of layers was set to 200 with a thickness of 1 cm. If not specified further, we left additional, unmentioned parameters at their default values.

Multicriterion Benchmark Test

To benchmark the model performance of IN^{3}M and MACRO, a multicriterion benchmarking strategy was applied based on multiple data sets: simulated vs. observed water content changes, solute distribution profiles, and maximum infiltration depths. Because IN^{3}M generates flow patterns of the simulated results, a direct qualitative comparison of the type, distribution, and depth of flow paths is possible. For this reason, the simulated vs. experimental flow patterns were checked for rough optical agreement in a first step. If the predicted IN^{3}M flow patterns completely differed from the observed ones (e.g., simulation of matrix flow only instead of deep-reaching preferential flow paths), no further quantitative benchmarking was undertaken.

As quantitative criteria for benchmarking both models, the following data sets were compared (see Fig. 1):

Table 3. Input parameters: macropore properties† for IN^{3}M and initial soil moisture of the five experimental sites‡ for IN^{3}M and MACRO. The measurements are assumed to be representative for a depth range of 0 to 15 cm (probe installed at the 10-cm depth), 15 to 25 cm (probe at the 20-cm depth), 25 to 35 cm (probe at the 30-cm depth), 35 to 45 cm (probe at the 40-cm depth), and 55 to 100 cm (probe at the 60-cm depth). For the depth range 45 to 55 cm, values were linearly interpolated.

Depth	NT			T			G1			G2			F		
	r_{mac}	n_{mac}	θ_{ini}	r_{mac}	n_{mac}	θ_{ini}	r_{mac}	n_{mac}	θ_{ini}	r_{mac}	n_{mac}	θ_{ini}	r_{mac}	n_{mac}	θ_{ini}
cm	mm		cm^3 cm^{-3}	mm		cm^3 cm^{-3}	mm		cm^3 cm^{-3}	mm		cm^3 cm^{-3}	mm		cm^3 cm^{-3}
0–5	2.09	72	0.153	1.13	3081	0.147	1.45	161	0.219	2.45	81	0.292	1.69	79	0.217
5–10	2.57	52	0.153	1.13	3081	0.147	1.45	161	0.219	2.45	81	0.292	1.69	79	0.217
10–15	1.91	88	0.153	1.13	3081	0.147	1.45	161	0.219	2.45	81	0.292	1.84	92	0.217
15–20	2.2	79	0.183	1.13	3407	0.183	2.22	104	0.216	4.59	68	0.280	1.84	92	0.201
20–25	2.2	79	0.183	1.13	3407	0.183	2.22	104	0.216	4.59	68	0.280	2.07	84	0.201
25–30	2.2	79	0.158	1.13	3407	0.163	1.63	156	0.229	4.59	68	0.225	2.07	84	0.187
30–35	2.34	60	0.158	2.01	249	0.163	1.63	156	0.229	3.14	80	0.225	2.07	84	0.187
35–40	2.34	60	0.095	2.01	249	0.143	1.63	156	0.224	3.14	80	0.159	2.07	84	0.150
40–45	2.34	60	0.095	2.01	249	0.143	1.63	219	0.224	3.14	80	0.159	2.08	68	0.150
45–50	2.79	26	0.091	2.29	79	0.167	1.63	219	0.235	4.02	47	0.159	2.08	68	0.127
50–55	2.79	26	0.091	2.29	79	0.167	1.63	219	0.235	4.02	47	0.159	2.08	68	0.127
55–60	2.79	26	0.088	2.29	79	0.190	1.59	152	0.247	4.02	47	0.341	2.08	68	0.105
60–65	2.79	26	0.088	2.29	79	0.190	1.59	152	0.247	3.08	29	0.341	2.08	68	0.105
65–70	2.79	26	0.088	2.66	71	0.190	1.59	152	0.247	3.08	29	0.341	2.08	68	0.105
70–75	2.79	26	0.088	2.66	71	0.190	1.59	152	0.247	3.08	29	0.341	2.08	68	0.105
75–80	2.79	26	0.088	2.66	71	0.190	1.59	152	0.247	3.08	29	0.341	2.08	68	0.105
80–85	2.79	26	0.088	2.66	71	0.190	1.59	152	0.247	3.08	29	0.341	2.08	68	0.105
85–90	2.79	26	0.088	2.66	71	0.190	1.59	152	0.247	3.08	29	0.341	2.08	68	0.105
90–95	2.79	26	0.088	2.66	71	0.190	1.59	152	0.247	3.08	29	0.341	2.08	68	0.105
95–100	2.79	26	0.088	2.66	71	0.190	1.59	152	0.247	3.08	29	0.341	2.08	68	0.105

† r_{mac}, macropore radius; n_{mac}, number of macropores; θ_{ini}, initial soil moisture contents (mean of two time domain reflectometry measurements per depth).
‡ NT, untilled farmland; T, tilled farmland; G, grassland; F, forest.

- Criteria 1 and 2: Depth profiles of relative water content changes after 75 and 125% of the sprinkling time, respectively. A depth profile consists of five measurements in the 10-, 20-, 30-, 40-, and 60-cm depths. The observed water content changes represent the mean of two TDR measurements per depth. Simulated water content changes represent the mean of a 10-cm-thick layer per measurement depth.

- Criterion 3: Depth profiles of solute distribution 24 h after the start of sprinkling. For IN^{3}M, simulated vs. observed depth profiles of the mean volume density (the approximate percentage of stained soil volume per depth increment) were compared. The mean volume density refers to 7 to 10 experimental dye patterns and seven simulated ones. For MACRO, simulated depth profiles of solute concentration were compared with the observed depth profiles of the mean volume density. For this reason, experimental volume density profiles had to be converted into concentration curves (solute concentration per depth increment). The conversion was based on the assumption that stained soil in the images exhibits a tracer concentration of 4 g L^{-1}.

- Criterion 4: Maximum infiltration depth. Observed and simulated (IN^{3}M) maximum infiltration depth refers to the highest value in one of the 7 to 10 experimental and seven simulated dye patterns, respectively, representing 5% volume density. The maximum infiltration depth of the MACRO simulations was derived from the solute concentration depth profiles. Because the measurements were only made to the 100-cm depth, we only considered the top 100 cm for extracting the simulated maximum infiltration depth.

Table 4. Input parameters: macropore parameters† for MACRO for three different parameterization cases (A–C)‡.

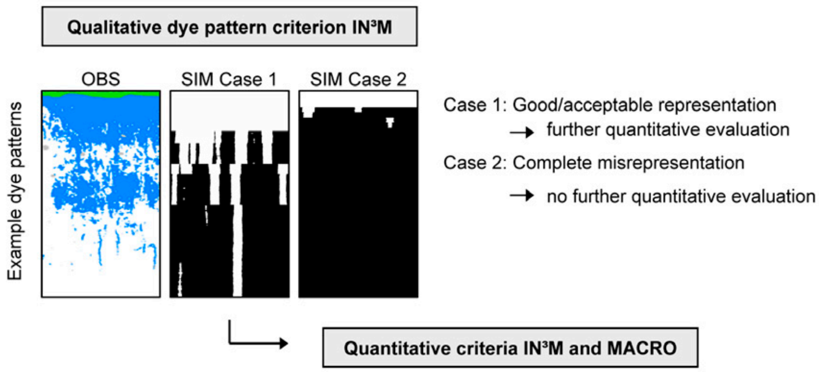
			TPORV			KSATMIN		ASCALE		CTEN	
Site§	Horizon	Depth	A	B	C	A	B & C	A	B & C	A	B & C
		cm				—— mm h^{-1} ——		—— mm ——		—— cm ——	
NT	Ah1	5–10	0.600	0.531	0.471	191.81	216,000	150	33	10	1
	Ah2	10–30	0.456	0.410	0.377	139.31	216,000	150	25	10	1
	E	30–40	0.424	0.382	0.357	473.51	216,000	150	36	10	1
	Bt-C	40–75	0.390	0.318	0.355	84.18	216,000	150	52	10	1
	C	75–100	0.353	0.338	0.313	76.43	216,000	150	52	10	1
T	Ah	0–30	0.598	0.556	0.475	566.14	216,000	150	4	10	1
	E1	30–40	0.351	0.339	0.308	53.00	216,000	150	35	10	1
	E2	40–65	0.318	0.308	0.296	54.83	216,000	150	26	10	1
	Bt-C	65–100	0.315	0.309	0.296	8.80	216,000	150	28	10	1
G1	Ah/E1	0–40	0.349	0.331	0.314	65.64	216,000	150	24	10	1
	E2	40–70	0.318	0.309	0.295	32.03	216,000	150	20	10	1
	Bt-C	70–100	0.315	0.307	0.293	8.78	216,000	150	19	10	1
G2	Ah	0–20	0.525	0.475	0.406	143.91	216,000	150	30	10	1
	Bw1	20–30	0.494	0.459	0.380	163.36	216,000	150	25	10	1
	Bw2	30–55	0.496	0.433	0.422	74.58	216,000	150	33	10	1
	Bg/Cr	55–65	0.460	0.433	0.382	166.74	216,000	150	57	10	1
	Bg/Cr	65–80	0.496	0.443	0.394	236.09	216,000	150	57	10	1
	Bg/Cr	80–100	0.458	0.428	0.366	202.68	216,000	150	57	10	1
F	Ah	0–10	0.584	0.532	0.448	94.84	216,000	150	31	10	1
	Bw1	10–15	0.563	0.534	0.455	130.67	216,000	150	29	10	1
	Bw2	15–25	0.529	0.488	0.410	167.36	216,000	150	29	10	1
	BwC	25–55	0.529	0.494	0.412	115.81	216,000	150	35	10	1
	C	55–100	0.568	0.528	0.430	137.06	216,000	150	35	10	1

† TPORV, saturated water content of micropores and macropores, calculated from saturated water content of micropores plus macroporosity (macroporosity = macropore area per horizontal soil section); KSATMIN, saturated hydraulic conductivity of macropores, calculated from the range of macropore flow rates measured by Weiler (2001) (3 cm^3 s^{-1}, 0.5-cm^2 macropore area); ASCALE, effective diffusion path length: the half-distance between macropores (distance refers to nearest neighbor distance assuming a random pattern of macropores); CTEN, capillary rise in macropores, calculated with the measured macropore radius.

‡ A: micropore and macropore parameters estimated with the MACRO pedotransfer function (PTF), macropore parameters based on soil structure properties representing a strong aggregate structure (structure: coarse, strength: strong, shape: prismatic); B: micropore parameters estimated with the MACRO PTF based on soil texture and bulk density, macropore parameters derived from experimental data; C: micropore parameters estimated with the ROSETTA PTF, macropore parameters derived from experimental data.

§ NT, untilled farmland; T, tilled farmland; G, grassland; F, forest.

As a goodness-of-fit measure, the mean absolute error (MAE) was calculated for Criteria 1 to 3. The mean error (ME) was determined for the maximum infiltration depth (simulated minus observed maximum infiltration depths). This choice of goodness-of-fit measures over commonly used measures such as the Nash–Sutcliffe model efficiency was made to keep the units and thus be able to better assess the magnitude of deviation and, for the maximum infiltration depth, also the direction. The goodness-of-fit of each criterion was than classified as good, acceptable, or failed based on an arbitrarily chosen threshold of the maximum MAE/ME (see Fig. 1). The maximum MAE of Criteria 1 to 3 was calculated by assuming a complete misrepresentation of simulated vs. observed data, e.g., high water content changes solely in the topsoil resulting from homogeneous matrix flow vs. low changes throughout the whole depth profile due to deep preferential flow. This resulted in a maximum MAE of 10% for the water content change criteria. The maximum ME of the maximum infiltration depth was set to 50 cm. Up to 20% deviation from the maximum MAE/ME was classified as good, up to 30% as acceptable, >30% as failed. In total, at least three of the four criteria had to be classified as good or acceptable to be an overall acceptable model performance.



No.	1	2	3		4
Criterion	Depth profile of relative water content changes after 75% of the sprinkling time	Depth profile of relative water content changes after 125% of the sprinkling time	Depth profile of solute distribution 24 h after start of sprinkling		Maximum infiltration depth
Description	Depth profile of five measurements in 10, 20, 30, 40 and 60 cm depth.		IN³M: Depth profile of mean volume density 1)	MACRO: Depth profile of solute concentration	Maximum value derived from depth profiles of volume density or solute concentration
Max. MAE/ME	10 %	10 %	1 [-]	4 g L^{-1}	50 cm
Classification of model performance	Good: < 20% of MAE/ME Acceptable: < 30% of MAE/ME Failed: > 30% of MAE/ME				

1) IN³M: Volume density = percentage of stained soil volume per depth increment. Mean volume density refers to 7-10 experimental dye patterns and 7 simulated ones.

Fig. 1. Multicriterion benchmark test; MAE = mean absolute error, ME = mean error.

Results

Simulations with IN^{3}M

As can be seen in Table 5, only the grassland sites passed the qualitative dye pattern criterion. The flow patterns of the forest and farmland sites were poorly simulated for either way of parameterization. Because the shape, depth, and distribution of flow paths completely differed from the experimental ones at the tilled, untilled, and forest sites, the model performance was classified as failed and no further benchmark criteria calculated. The quantitative benchmark test for both grassland sites led to the following picture (see Table 6): The flow processes in one grassland site (G1) were predominantly well captured by IN^{3}M. At the other grassland site (G2), poor model fits were obtained for all criteria except for the volume density profiles.

The predicted vs. experimental volume density profiles of G1 and G2 are displayed in Fig. 2 (right panel). Although the depth of topsoil matrix staining was underestimated at both grassland sites (except for parameterization with the ROSETTA PTF at G2), the general trend of solute distribution was reasonably represented. The increase in volume density at about the 50-cm depth at both grassland sites resulted from lateral dye spreading due to an impeding layer of higher clay content. In Fig. 2 on the left, the results of the relative water content changes after 125% of the sprinkling time are shown. Soil moisture changes in G1 were better captured

Table 5. IN^{3}M simulations: qualitative dye pattern criteria for the five experimental sites†.

Site	NT		T		G1		G2		F	
Parameterization	1	2	1	2	1	2	1	2	1	2
Classification‡	f	f	f	f	a	a	a	a	f	f

† NT, untilled farmland; T, tilled farmland; G, grassland; F, forest.
‡ Classification: g = good, a = acceptable, f = failed simulation.

Table 6. The IN^{3}M simulations: quantitative criteria for the two grassland sites (G1 and G2). Good simulations are in bold, acceptable in bold italic; failed simulations are unmarked. The 20% threshold of mean absolute error (MAE)/mean error (ME): Criterion 1 and 2, 2%; Criterion 3, 0.2; Criterion 4, 10 cm. The 30% threshold of MAE/ME: Criteria 1 and 2, 3%; Criterion 3, 0.3; Criterion 4, 15 cm. At least three of the four criteria have to be classified as acceptable for an overall acceptable model performance.

Site	Parameterization	Criterion No. 1	Criterion No. 2	Criterion No. 3	Criterion No. 4	Overall classification†
		%	%		cm	
G1	1	**1.72**	3.31	**0.14**	***13***	a
	2	***2.13***	***2.56***	**0.15**	***13***	a
G2	1	3.42	8.11	***0.21***	−39	f
	2	5.35	6.42	**0.12**	31	f

† Overall classification: g = good, a = acceptable, f = failed.

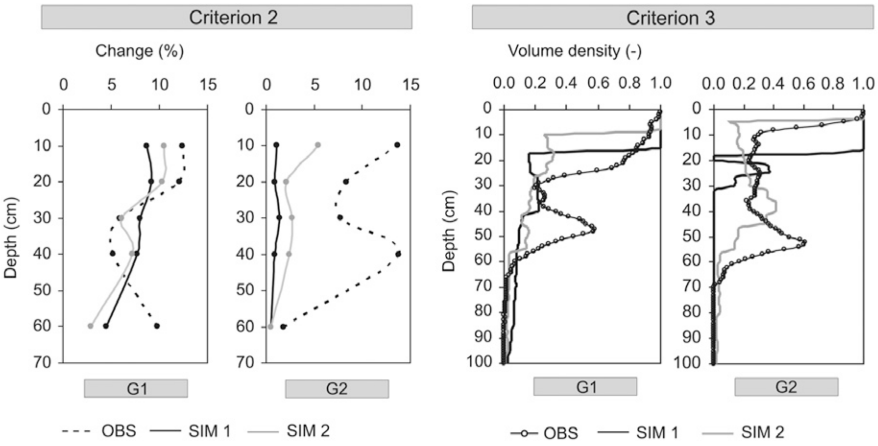


Fig. 2. Example of IN^{3}M simulation results for the two grassland sites (G1 and G2): Criterion 2 is the relative water content change after 125% of the sprinkling time (300 min); Criterion 3 is the depth profiles of volume density, which refers to the mean volume density averaged across the 7 to 10 dye patterns; OBS = observed data; SIM 1 and SIM 2 = simulation parameterizations 1 and 2.

than in G2, even though being above and close to the threshold of an acceptable simulation for parameterization with the ROSETTA PTF and the MACRO PTF, respectively. The poor model fit of soil moisture changes in G2 can be partly explained by the mentioned effect of an impeding layer, which caused lateral diversion and high water content changes at roughly the 50-cm depth. Additionally, the simulated low changes in soil moisture resulted from flow routing to great depths. This was revealed by the qualitative dye patterns, where simulated flow paths reached >100 cm in both grassland sites. Despite a simulation depth of 200 cm, however, only the top 100 cm was considered for extracting the maximum infiltration depth. Overall, the maximum infiltration depth in G1 was overestimated. For G2, parameterization with the ROSETTA PTF also led to deeper flow than observed. In contrast, parameterization with the MACRO PTF caused an underestimation of flow depth (see Table 6).

Simulations with MACRO

The simulation results obtained with the MACRO model were quite similar to those with IN^{3}M regarding the overall model ability to reproduce flow processes. As shown in Table 7, the agreement between simulations and observations was generally poor for all sites except G1. Even though some of the criteria were well or acceptably met for the forest, G2, and farmland sites, the overall model performance was classified as failed. Only G1 was well described by the MACRO model.

Looking at each benchmark criterion in more detail, the water content changes after 75 and 125% of the sprinkling time were well reproduced or exceeded the acceptance limit only marginally at G1 and the untilled farmland (NT) for most parameterization cases (see Table 7). In contrast, the simulation results of the tilled farmland (T), forest (F), and G2 were far above the threshold of an acceptable simulation for most parameterizations. Interestingly, the benchmark criteria of solute distribution and maximum infiltration depth showed contrasting results for the untilled and forest sites regarding the quality of model fit. In the case of the untilled field, a benchmark solely based on water content changes yields a good fit. Through the additional consideration of solute distribution profiles and maximum infiltration depth, this good agreement is thwarted. For the forest site, the solute distribution criterion leads to the impression of the model being capable of reproducing flow processes. When the water content change criterion is incorporated into the benchmarking, however, this conclusion is disqualified.

This discrepancy of results depending on the considered benchmark criteria can be observed in Fig. 3. Here, two examples of relative water content changes after 125% of the sprinkling time (left panel) and solute distribution profiles (right panel) are plotted. Because MACRO is frequently used for predictions in agricultural soil, it was chosen

Table 7. The MACRO simulations: quantitative criteria. Good simulations are in bold, acceptable in bold italic; failed simulations are unmarked. The 20% threshold of mean absolute error (MAE)/mean error (ME): Criteria 1 and 2, 2%; Criterion 3, 0.8 g L^{-1}; Criterion 4, 10 cm. The 30% threshold of MAE/ME: Criteria 1 and 2, 3%; Criterion 3, 1.2 g L^{-1}; Criterion 4: 15 cm. At least three of the four criteria have to be classified as acceptable for an overall acceptable model performance.

Site†	Parameterization	Criterion No. 1	No. 2	No. 3	No. 4	Overall classification‡
		—— %	——	g L^{-1}	cm	
NT	A	**1.43**	**1.20**	1.54	−37	f
	B	**0.83**	**1.64**	1.77	−43	f
	C	***2.54***	3.69	1.70	−38	f
T	A	5.30	9.43	2.53	−43	f
	B	5.38	9.84	2.41	52	f
	C	**0.47**	5.46	2.53	−43	f
G1	A	***2.25***	3.44	***1.07***	**−10**	a
	B	***2.67***	3.20	***1.02***	***−13***	a
	C	***2.64***	3.51	***1.20***	***13***	a
G2	A	***2.48***	4.37	2.41§	−19	f
	B	4.97	6.52	2.19	−25	f
	C	3.54	***2.23***	2.62	−38	f
F	A	7.71	7.50	*0.68*	**−8**	f
	B	10.31	10.52	***0.84***	−18	f
	C	4.40	5.11	1.87	17	f

† NT, untilled farmland; T, tilled farmland; G, grassland; F, forest.
‡ Overall classification: g = good, a = acceptable, f = failed.
§ Solute balance error of 25%.

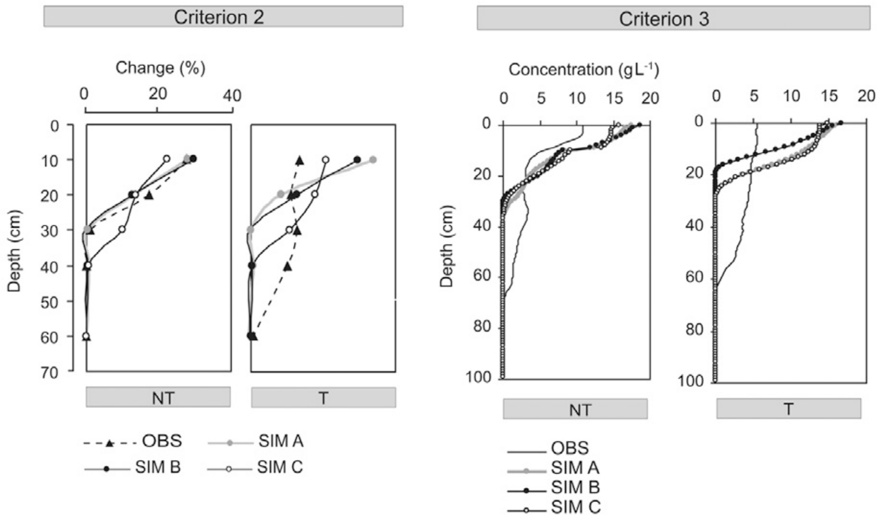


Fig. 3. Examples of MACRO simulation results for the two agricultural sites (NT, untilled; T, tilled): Criterion 2 is the relative water content changes after 125% of the sprinkling time (300 min); Criterion 3 is the depth profiles of volume density, which refers to the mean volume density averaged across the 7 to 10 dye patterns; OBS = observed data; SIM A, SIM B, and SIM C = simulation parameterizations A, B, and C.

to depict the results of both farmland sites. The observed soil moisture depth profile in the untilled field (high water content changes in the topsoil, no changes below 30 cm) was well imitated by the simulations parameterized according to Cases A and B. Parameterization Case C led to lower values in the topsoil and larger changes farther down in the profile than measured. The corresponding solute distribution profile of the NT site (Fig. 2, right panel) shows the discrepancy of results for the benchmark Criteria 1 and 2 (water content) vs. Criteria 3 and 4 (solute distribution and maximum infiltration depth): solute transport to roughly the 60-cm depth was observed in both agricultural soils. At the NT site, this contradicts the process information revealed by the TDR measurements, where water content changes were observed in just the topsoil. Because TDR measurements include only a small soil volume, however, spatially highly variable flow to a great depth, as depicted in the dye patterns, was not covered.

In the tilled field, the observed soil moisture changes and solute transport up to the 60-cm depth were not captured by MACRO (see Fig. 3). Instead, higher water content changes in the topsoil were simulated. An underestimation of soil moisture changes at greater depths at the expense of higher water content changes in the topsoil were also observed for G1, G2, and F for most parameterizations (graphs not displayed). To highlight the effects of different parameterizations, Case C at both farmland sites resulted in higher soil moisture changes farther down in the profile than Cases A or B after 75 and 125% of the sprinkling time. This also applies to the forest site (graphs not shown). At F and G1, Case C led also to a higher maximum infiltration depth than Cases A and B. Because this is not consistent at all sites, however, a clear effect of the different parameterizations on the simulations is not inferable.

To summarize the outcome of both model simulations, IN^{3}M and MACRO showed a similar overall ability or inability to capture flow processes in soils developed under different LULCs. Only flow processes in one grassland site (G1) could be reproduced. The IN^{3}M and MACRO models differed, however, in terms of internal processes. As shown, IN^{3}M overestimated the maximum infiltration depth at both grassland sites (except for parameterization with the ROSETTA PTF at G2). To the contrary, MACRO underestimated the solute transport depth.

Discussion

Representation of Land Use Effects

As revealed by the multicriterion benchmark test, both models designed for water and solute transport in structured soil showed only a limited capability to reproduce the observed data. Flow processes in all sites except one of the grassland sites (G1) were poorly simulated. This is especially relevant because MACRO is a widely used model for predicting pesticide leaching in agricultural soil. From the benchmark test, it can be inferred that both models exhibit several structural shortcomings with regard to land use effects. Nevertheless, it has to be emphasized that it is difficult to differentiate between the effect of model structure error and parameter uncertainty. Because the model structure and the PTFs were evaluated at the same time, the simulation weaknesses cannot be clearly attributed to model structure deficiencies alone; however, certain discrepancies between simulations and observations appeared independent of the different parameterizations. Thus, the following explanations for the poor model fit due to LULC effects are suggested.

The initiation of preferential flow due to a strong surface microtopography at both farmland sites and water repellency of the litter layer in the forest was insufficiently represented by IN^{3}M or MACRO. At both agricultural sites, irregularly spaced microdepressions in the untilled site and a strong ridge-and-furrow surface topography in the tilled field caused water accumulation and infiltration only in the depressions. Hence, irrigation water was diverted into only a small proportion of the macropores—those located in the concavities. Particularly in the tilled field, infiltration was initiated in the furrows only. This spatially variable water supply to macropores at the surface was also detected by Weiler and Naef (2003b) and thus considered in the representation of macropore flow initiation in IN^{3}M. The inflow quantity of each macropore in IN^{3}M is calculated according to a probability distribution function of the MDA. Yet, as Weiler and Naef (2003b) discussed in their study, the actual MDA distribution for other types of macropores or soil surface characteristics can be different than the derived distributions. Adapting these MDA distribution functions to different surface characteristics may be one step toward a better representation of the observed processes. Because MACRO does not consider this variable initiation, all macropores evenly participate in flow routing. This may explain the observed underestimation of flow depth by MACRO.

The description of structural features in IN^{3}M and MACRO simplifies the complexity of the macropore geometry, continuity, and spatial dimension observed in soils developed under different LULCs. An abundance of small interaggregate tillage voids (low-continuity, slab geometry) were found in the topsoil of the plowed study site. Because IN^{3}M only accounts for high-continuity earthworm burrows, flow in this type of pore was not adequately captured. Additionally, lateral roots strongly governed preferential flow in the forest site; however, macropores in both models are solely vertically oriented. Generally, the conceptualization of the macropore domain in the MACRO model is restricted to interaggregate macropores. Other voids such as biopores with different geometry, continuity, and spatial dimension are not conceptualized. Even though we adjusted the macropore parameters to represent the observed features (Cases B and C), the simulations were poor for most sites. Nonexperts using the internal PTF to estimate macropore parameters (as in Case A) are not able to account for pores other than interaggregate pores. A sandy, weakly aggregated soil interspersed with numerous biopores, for instance, cannot be adequately represented by solely accounting for interaggregate pores. The choice of aggregate structure in Case A (strong, prismatic aggregate structure) is an unrealistic choice for the study sites (soil texture dominantly silty sand and sandy silt). Nonetheless, it was chosen to mimic structural features of high continuity. With this choice of strong soil structure, the solute transport was underestimated for all sites. Similar findings were reported by Scorza et al. (2007), who tested MACRO 5.1 against experimental leaching data of a nonreactive tracer (Br^-) and two pesticides in a cracked clay soil. They first calibrated MACRO against measured moisture profiles and Br^- concentrations in the soil and the drainage water. Uncalibrated pesticide simulations were then compared with field data on pesticide leaching. The calibrated parameter values revealed that a high degree of physical nonequilibrium (i.e., strong macropore flow) was necessary to describe solute transport in this soil (Scorza et al., 2007).

Apart from the above LULC effects, the following reasons beyond parameter uncertainty may also be responsible for the discrepancy between simulations and observations. First, both models omit other types of preferential flow mechanisms (unstable or funnel flow). As the detailed analysis of field data by Bachmair et al. (2009) revealed, however, unstable flow occurred in the untilled field and funnel flow in the tilled and both grassland sites. In the grassland sites, funnel flow resulted from an impeding layer of higher clay content causing lateral diversion. In the tilled field, the lateral spreading can be ascribed to the plow pan. Second, an unsatisfactory description of the interaction process (water flow from the macropores into the surrounding soil matrix) may have introduced errors. This was already pointed out by Larsson and Jarvis (1999), who evaluated the MACRO model with field data from a 1-yr leaching experiment in a structured clay soil. They concluded that one source of model error may be the simplified description of first-order mass exchange between micro- and macropores, which neglects lateral concentration gradients. For IN^{3}M, the over- and underestimation of flow depth could also result from additional factors governing the interaction process, e.g., macropore coatings or organic matter content (Gerke, 2006; Jarvis, 2007), which, in turn, will most likely be influenced by land use.

From the shortcomings highlighted above, it can be inferred that the conceptualization and parameterization of mechanisms governing infiltration and percolation in IN^{3}M and MACRO are insufficient for predictive simulations of flow and transport in soils under different LULCs. Our benchmark test revealed that the effects of LULC cannot be captured through simply changing the parameters linked to LULC, such as adjusting the amount, size, and spatial distribution of macropores. Additional processes need to be conceptualized, e.g., horizontal flow along roots or the effect of variable inflow into macropores due to a strong surface microtopography in agricultural soils or surface water repellency in forest soils. This model structural error is often masked through model calibration. To predict the impact of land use change on water resources, changes in soil properties resulting from land use have to be considered. The necessity of accounting for LULC effects is also relevant for larger scale hydrologic models, where concepts of land-use-driven processes have also been rarely implemented (Niehoff et al., 2002).

Choice of Benchmarking Strategy

The applied benchmark test based on multiple criteria served as a helpful tool to assess the capability of IN^{3}M and MACRO to simulate water flow and solute transport in heterogeneous soil under different LULC. Each benchmark criterion by itself has potential drawbacks. When considered together, however, a comprehensive picture of the overall model performance was gained. The necessity of using multiple criteria for a reliable model benchmark was illustrated for the MACRO simulation results of the untilled farmland site. A good fit of simulated vs. observed depth profiles of relative water content changes was achieved. Nonetheless, through the additional consideration of solute distribution profiles and maximum infiltration depth, this good agreement was thwarted. This results from the nature of TDR measurements, which are solely "snapshots" of space in time, not representative of a larger soil volume (Jarvis et al., 2000). Apart from the need to compensate for flaws in the experimental data, multiple data sets are essential for examining the adequacy of internal model behavior. For instance, IN^{3}M simulated low overall water content changes in the first grassland site, which fitted the observed data well. Yet these simulated low values resulted from flow routing to great depth. As the experimental dye patterns revealed, solute transport to this depth did not occur in reality. This example highlights the insufficient internal process description of IN^{3}M. We therefore urge the application of multicriterion benchmarking strategies combining information of high spatial resolution but limited temporal dimension (e.g., dye patterns) with information of low spatial but high temporal precision (e.g., TDR measurements) to fully capture model strengths and weaknesses. This necessity of using multiple criteria for model validation purposes has also been stressed by others (Seibert and McDonnell, 2002; Bathurst et al., 2004).

What has to be kept in mind when interpreting the results of the benchmark test is the choice of acceptance criteria. A benchmark is always arbitrary. The calculation of the maximum MAE and maximum ME by assuming a complete misrepresentation of simulated vs. observed data is also somewhat subjective; however, the resulting goodness-of-fit measure for each simulation criterion per model run was presented. Hence, the impact of increasing or decreasing the threshold of an acceptable simulation can be easily assessed. We found that if the benchmark was exceeded, it was mostly a strong deviation indicating a clear misfit due to the various reasons discussed above.

The chosen goodness-of-fit measure (MAE/ME) also helped to judge the appropriateness of the modeled processes because it provides information on the magnitude and direction of deviation in the measured units. In this context, the value of qualitative data as benchmark criteria should be emphasized. In our study, the qualitative comparison of simulated vs. observed dye patterns was an effective criterion for roughly judging the predictive power of IN^{3}M. Without having to calculate quantitative criteria, the user can directly apply his or her process knowledge to evaluate the data. The artificial dye patterns in IN^{3}M thus provide a fast and intuitive way to assess the adequacy of modeled processes. Particularly for the verification of new model concepts, this is a very effective method to gain process information (Weiler, 2005).

Conclusions

To benchmark the predictive capacity of two dual-permeability models to correctly simulate flow and transport in structured soil under different LULCs, sprinkling experiments were conducted on five sites with similar soil texture, yet differences in LULC. Multiple data sets were collected to benchmark the simulation results (water content changes at different depths, solute distribution profiles, and maximum infiltration depth). Both models were partially capable of predicting water flow and solute transport in one grassland site. Flow processes at all other sites (tilled and untilled farmland, deciduous forest, and a second grassland site) were poorly simulated. Among other factors, these weaknesses may be ascribed to additional flow-governing processes not included into the model structure. Poor model outcomes were attributed to LULC effects such as a strong surface microtopography altering macropore flow initiation in both agricultural soils or horizontally oriented roots and surface water repellency in the forest soil. In conclusion, this suggests that LULC effects need to be better incorporated into the conceptualization and parameterization of infiltration and percolation in hydrologic models to obtain realistic predictions concerning water quality and quantity. We need more studies benchmarking the predictive capacity of models because this provides us with the possibility to improve our models and to include important processes governing flow in the unsaturated zone. Moreover, we emphasize the need to apply benchmarking strategies based on multiple criteria to fully capture model strengths and weaknesses.

References

Bachmair, S., M. Weiler, and G. Nützmann. 2009. Controls of land use and soil structure on water movement: Lessons for pollutant transfer through the unsaturated zone. J. Hydrol. 369:241–252.

Bathurst, J., J. Ewen, G. Parkin, P. O'Connell, and J. Cooper. 2004. Validation of catchment models for predicting land-use and climate change impacts: 3. Blind validation for internal and outlet responses. J. Hydrol. 287:74–94.

Beven, K. 2001. On explanatory depth and predictive power. Hydrol. Processes 15:3069–3072.

Bodhinayake, W., and B. Si. 2004. Near-saturated surface soil hydraulic properties under different land uses in the St Denis National Wildlife Area, Saskatchewan, Canada. Hydrol. Processes 18:2835–2850.

Bogner, C., B. Wolf, M. Schlather, and B. Huwe. 2008. Analysing flow patterns from dye tracer experiments in a forest soil using extreme value statistics. Eur. J. Soil Sci. 59:103–113.

Bormann, H., and K. Klaassen. 2008. Seasonal and land use dependent variability of soil hydraulic and soil hydrological properties of two northern German soils. Geoderma 145:295–302.

Dubus, I., and C. Brown. 2002. Sensitivity and first-step uncertainty analyses for the preferential flow model MACRO. J. Environ. Qual. 31:227–240.

Dubus, I., C. Brown, and S. Beulke. 2003. Sources of uncertainty in pesticide fate modelling. Sci. Total Environ. 317:53–72.

Edwards, W.M., L.D. Norton, and C.E. Redmond. 1988. Characterizing macropores that affect infiltration into nontilled soil. Soil Sci. Soc. Am. J. 52:483–487.

Ewen, J., and G. Parkin. 1996. Validation of catchment models for predicting land-use and climate change impacts: 1. Method. J. Hydrol. 175:583–594.

Flury, M., H. Flühler, W. Jury, and J. Leuenberger. 1994. Susceptibility of soils to preferential flow of water: A field study. Water Resour. Res. 30:1945–1954.

Gerke, H. 2006. Preferential flow descriptions for structured soils. J. Plant Nutr. 169:382–400.

Hendrickx, J.M.H., and M. Flury. 2001. Uniform and preferential flow mechanisms in the vadose zone. p. 149–187. *In* National Research Council (ed.) Conceptual models of flow and transport in the fractured vadose zone. Natl. Acad. Press, Washington, DC.

Horn, R., H. Taubner, M. Wuttke, and T. Baumgartl. 1994. Soil physical properties related to soil structure. Soil Tillage Res. 30:187–216.

Jarvis, N. 2007. A review of non-equilibrium water flow and solute transport in soil macropores: Principles, controlling factors and consequences for water quality. Eur. J. Soil Sci. 58:523–546.

Jarvis, N., C. Brown, and E. Granitza. 2000. Sources of error in model predictions of pesticide leaching: A case study using the MACRO model. Agric. Water Manage. 44:247–262.

Jarvis, N., P. Jansson, P. Dik, and I. Messing. 1991. Modelling water and solute transport in macroporous soil: I. Model description and sensitivity analysis. Eur. J. Soil Sci. 42:59–70.

Kasteel, R., M. Burkhardt, S. Giesa, and H. Vereecken. 2005. Characterization of field tracer transport using high-resolution images. Vadose Zone J. 4:101–111.

Kirchner, J. 2006. Getting the right answers for the right reasons: Linking measurements, analyses, and models to advance the science of hydrology. Water Resour. Res. 42:W03S04, doi:10.1029/2005WR004362.

Klemeš, V. 1986. Operational testing of hydrological simulation models. Hydrol. Sci. J. 31:13–24.

Larsbo, M., S. Roulier, F. Stenemo, R. Kasteel, and N. Jarvis. 2005. An improved dual-permeability model of water flow and solute transport in the vadose zone. Vadose Zone J. 4:398–406.

Larsson, M., and N. Jarvis. 1999. Evaluation of a dual-porosity model to predict field-scale solute transport in a macroporous soil. J. Hydrol. 215:153–171.

Niehoff, D., U. Fritsch, and A. Bronstert. 2002. Land-use impacts on storm-runoff generation: Scenarios of land-use change and simulation of hydrological response in a meso-scale catchment in SW-Germany. J. Hydrol. 267:80–93.

Öhrström, P., M. Persson, J. Albergel, P. Zante, S. Nasri, R. Berndtsson, and J. Olsson. 2002. Field-scale variation of preferential flow as indicated from dye coverage. J. Hydrol. 257:164–173.

Oswald, S., and W. Kinzelbach. 2004. Three-dimensional physical benchmark experiments to test variable-density flow models. J. Hydrol. 290:22–42.

Schaap, M., F. Leij, and M.Th. van Genuchten. 2001. ROSETTA: A computer program for estimating soil hydraulic parameters with hierarchical pedotransfer functions. J. Hydrol. 251:163–176.

Schwartz, R., S. Evett, and P. Unger. 2003. Soil hydraulic properties of cropland compared with reestablished and native grassland. Geoderma 116:47–60.

Scorza, R., Jr., N. Jarvis, J. Boesten, S. van der Zee, and S. Roulier. 2007. Testing MACRO (version 5.1) for pesticide leaching in a Dutch clay soil. Pest Manage. Sci. 63:1011–1025.

Seibert, J., and J. McDonnell. 2002. On the dialog between experimentalist and modeler in catchment hydrology: Use of soft data for multicriteria model calibration. Water Resour. Res. 38(11):1241, doi:10.1029/2001WR000978.

Shipitalo, M., W. Dick, and W. Edwards. 2000. Conservation tillage and macropore factors that affect water movement and the fate of chemicals. Soil Tillage Res. 53:167–183.

Sponagel, H. (ed.). 2005. Bodenkundliche Kartieranleitung. 5th ed. German Federal Inst. of Geosci. and Nat. Resour., Hannover.

Vanderborght, J., R. Kasteel, M. Herbst, M. Javaux, D. Thiery, M. Vanclooster, C. Mouvet, and H. Vereecken. 2005. A set of analytical benchmarks to test numerical models of flow and transport in soils. Vadose Zone J. 4:206–221.

Wahren, A., K. Feger, K. Schwärzel, A. Münch, and I. Dittrich. 2009. Bodenhydraulische Änderungen durch Aufforstung und deren Bedeutung bei der modellgestützten Bewertung des Klimawandeleinflusses auf Standortswasserhaushalt und Hochwasserentstehung.) *In* Soil, a finite resource: Annu. Meet. German Soil Sci. Soc. (DGB), Bonn. 5–13 Sept. 2009. Forschungszentrum Jülich, Jülich, Germany.

Weiler, M. 2001. Mechanisms controlling macropore flow during infiltration. Ph.D. diss. Swiss Federal Inst. of Technol., Zürich.

Weiler, M. 2005. An infiltration model based on flow variability in macropores: Development, sensitivity analysis and applications. J. Hydrol. 310:294–315.

Weiler, M., and H. Flühler. 2004. Inferring flow types from dye patterns in macroporous soils. Geoderma 120:137–153.

Weiler, M., and F. Naef. 2003a. An experimental tracer study of the role of macropores in infiltration in grassland soils. Hydrol. Processes 17:477–493.

Weiler, M., and F. Naef. 2003b. Simulating surface and subsurface initiation of macropore flow. J. Hydrol. 273:139–154.

Zhou, X., H. Lin, and E. White. 2008. Surface soil hydraulic properties in four soil series under different land uses and their temporal changes. Catena 73:180–188.